

Course Creator Platform

Comprehensive User Guide

Version 3.3.1 | December 2025

Course Creator Platform - User Guide

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Introduction

Welcome to the Course Creator Platform - a comprehensive learning management system designed for corporate training and educational institutions. This platform enables organizations to create, deliver, and track professional development courses with AI-powered assistance.

Key Features

- **AI-Powered Content Generation:** Create courses, quizzes, and labs with AI assistance
 - **Interactive Lab Environments:** Hands-on coding practice with real-time feedback
 - **Multi-Tenant Architecture:** Secure organization isolation and management
 - **RAG-Enhanced AI Assistant:** Context-aware help and tutoring
 - **Comprehensive Analytics:** Track learning progress and engagement
 - **Integration Ready:** Connect with Zoom, Slack, and enterprise systems
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Platform Overview

System Architecture

The platform is built on a modern microservices architecture, ensuring scalability, reliability, and maintainability.

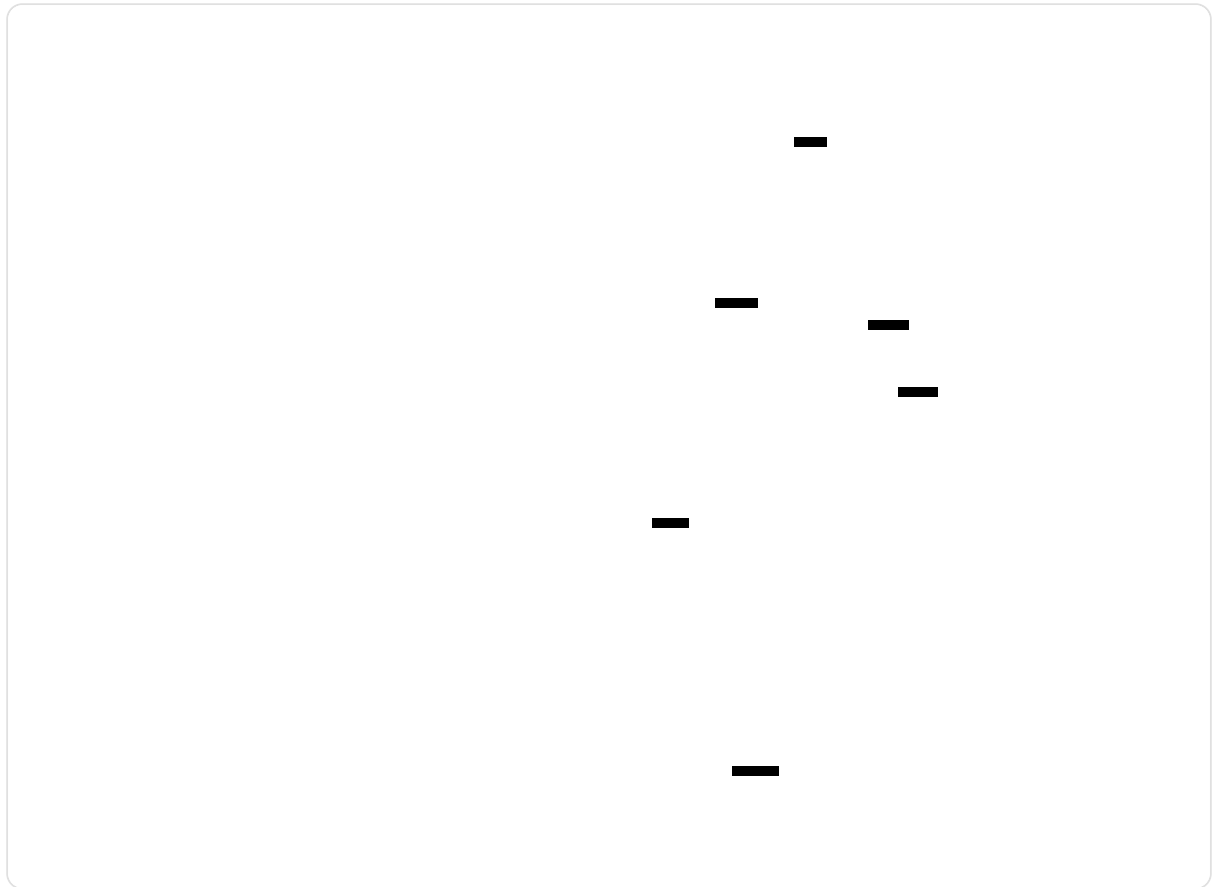


Key Components

Component	Description	Port
User Management	Authentication, authorization, RBAC	8000
Organization Management	Multi-tenant org management	8001
Course Management	Course CRUD, enrollment	8002
Content Management	Media, documents, resources	8003
Course Generator	AI content generation	8004
Lab Manager	Docker container orchestration	8005
Analytics	Learning analytics & reporting	8006
Metadata Service	Search, tags, categorization	8007
RAG Service	Vector search & retrieval	8009
AI Assistant	Conversational AI interface	8011

Data Flow

Understanding how data flows through the platform helps in troubleshooting and optimization.



Getting Started

System Requirements

For End Users:

- Modern web browser (Chrome, Firefox, Safari, Edge)
- Stable internet connection
- Screen resolution: 1280x720 or higher

For Lab Environments:

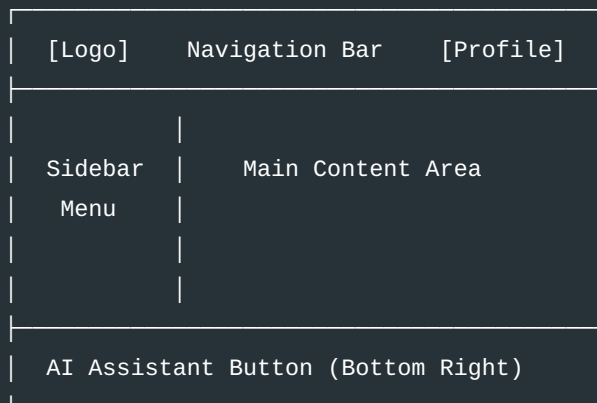
- WebSocket support in browser
- JavaScript enabled

First-Time Login

1. Navigate to the platform URL provided by your organization
2. Click "Login" or "Register" if creating a new account
3. Enter your credentials
4. Complete the AI Assistant welcome tour (recommended)

Navigation Overview

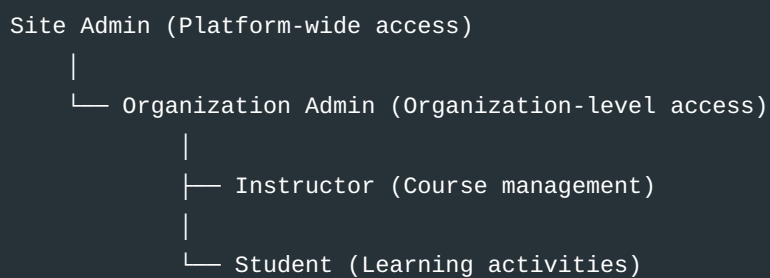
The platform uses a role-based dashboard system:



User Roles & Permissions

The platform supports five user roles with hierarchical permissions:

Role Hierarchy



|
└─ Guest (Limited public access)

Permission Matrix

Feature	Site Admin	Org Admin	Instructor	Student	Guest
View Public Courses					
Enroll in Courses					
Create Courses					
Manage Students					
Manage Instructors					
Organization Settings					
Platform Settings					
System Monitoring					

Student Guide

Student Workflow Overview

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Dashboard Features

Your student dashboard provides quick access to:

- **Enrolled Courses:** Current courses with progress indicators
- **Upcoming Deadlines:** Assignments, quizzes, and milestones
- **Recent Activity:** Continue where you left off
- **Certificates:** View and download earned certificates
- **Progress Overview:** Visual progress tracking

Taking Courses

Viewing Course Content

1. Navigate to **My Courses** from the dashboard
2. Click on a course to open it
3. Progress through modules sequentially or use the outline
4. Mark lessons as complete when finished

Interactive Learning Features

Video Lessons:

- Playback speed controls
- Caption support
- Bookmarking key moments
- Progress auto-save

Reading Materials:

- Downloadable resources
- Highlighted key concepts
- Searchable content

Lab Environments

Labs provide hands-on practice in isolated coding environments.

Starting a Lab

1. Navigate to the lab section within a course
2. Click "Start Lab Environment"
3. Wait for container initialization (10-30 seconds)
4. Write and run code in the integrated editor

Lab Features

- **Monaco Editor:** VS Code-like editing experience
- **Real-time Execution:** Run code and see output instantly
- **AI Assistance:** Get help from the AI assistant
- **Test Validation:** Automatic grading of solutions
- **Save Progress:** Work is auto-saved

Best Practices for Labs

- Read the instructions completely before starting
- Use the AI assistant for hints, not answers
- Test your code frequently
- Submit before the deadline

Quizzes & Assessments

Quiz Types

Type	Description	Feedback
Practice Quiz	Unlimited attempts, no grade impact	Immediate
Graded Quiz	Limited attempts, affects grade	After submission
Timed Quiz	Time limit, single attempt	After deadline

Taking a Quiz

1. Click "Start Quiz" from the course
2. Answer all questions
3. Review before submitting
4. View results and feedback

Earning Certificates

Certificates are awarded upon course completion meeting these criteria:

- All required modules completed
- Minimum quiz score achieved (typically 70%)
- All required labs submitted

Downloading Certificates

1. Go to **Certificates** from dashboard
 2. Click the certificate to preview
 3. Click "Download PDF" or "Share"
-

Instructor Guide

Instructor Workflow Overview

Dashboard Overview

The instructor dashboard provides:

- **My Courses:** Courses you teach
- **Student Activity:** Recent enrollments and submissions
- **Analytics:** Engagement metrics
- **Quick Actions:** Create course, view students

Creating Courses

Method 1: AI-Assisted Creation

1. Click **Create Course** → **Use AI Generator**
2. Enter course topic and objectives
3. Select target audience and duration
4. Click "Generate Outline"
5. Review and edit the generated outline
6. Click "Generate Content" for each section
7. Review, edit, and publish

Method 2: Manual Creation

1. Click **Create Course** → **Start from Scratch**
2. Fill in course details:
3. Title and description
4. Category and tags
5. Learning objectives
6. Prerequisites
7. Add modules and lessons
8. Upload or create content
9. Configure assessments
10. Preview and publish

Content Generation with AI

The AI content generator can create:

- **Course Outlines:** Structured curriculum
- **Lesson Content:** Text, slides, summaries
- **Quiz Questions:** Multiple choice, true/false, coding
- **Lab Exercises:** Code templates and test cases

Using the Content Generator

1. Navigate to: Instructor → Content Generator
2. Select content type
3. Provide context and requirements
4. Generate and review
5. Edit as needed
6. Add to course

Managing Students

Enrolling Students

Individual Enrollment:

1. Go to course → Students tab
2. Click "Enroll Student"
3. Search by name or email
4. Click "Enroll"

Bulk Enrollment:

1. Go to course → Students → Bulk Enroll
2. Download the template CSV
3. Fill in student information
4. Upload the completed CSV
5. Review and confirm

Tracking Progress

The student progress view shows:

- Completion percentage
- Quiz scores
- Lab submissions
- Time spent
- Last activity

Creating Labs

Lab Environment Setup

1. Navigate to course → Labs → Create Lab
2. Configure environment:
3. Programming language
4. Required packages
5. Memory/CPU limits
6. Create starter code template
7. Define test cases
8. Set grading criteria
9. Preview and publish

Analytics & Reporting

Access detailed analytics:

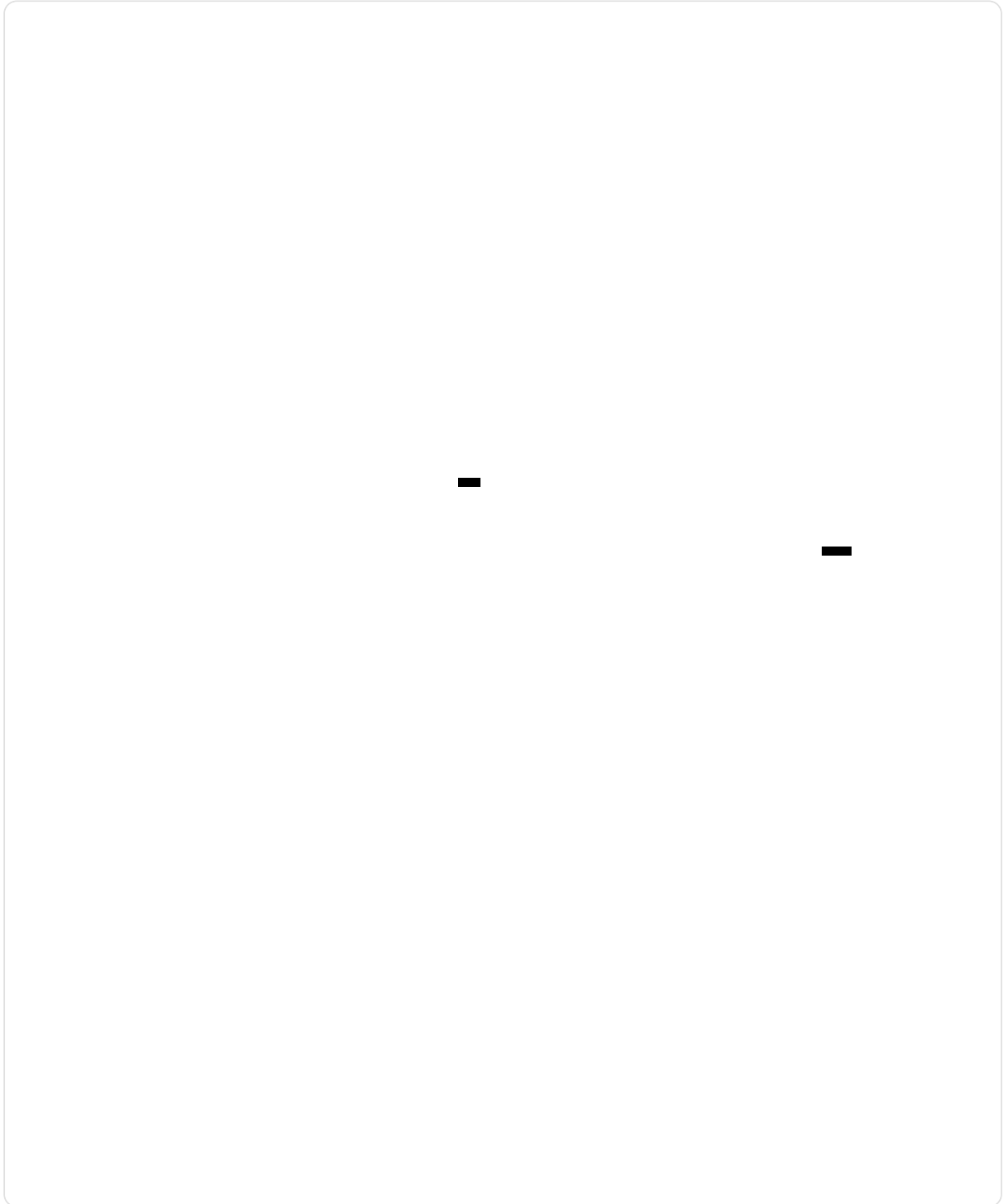
- **Engagement:** Time on platform, video completion
- **Performance:** Quiz scores, lab grades
- **Trends:** Progress over time
- **Comparisons:** Cohort analysis

Generating Reports

1. Go to Analytics → Reports
 2. Select report type
 3. Choose date range
 4. Select students/courses
 5. Export as PDF or CSV
-

Organization Admin Guide

Organization Admin Workflow



Organization Setup

Initial Configuration

1. **Organization Profile**

2. Name and description
3. Logo and branding
4. Contact information

5. **Compliance Settings**

6. Data retention policies
7. Privacy preferences
8. GDPR/CCPA compliance

9. **Integration Setup**

10. SSO configuration
11. Zoom integration
12. Slack notifications

Member Management

Inviting Members

1. Go to **Members** → **Invite**
2. Enter email address(es)
3. Select role (Student, Instructor)
4. Send invitation

Bulk Import

1. Download member template
2. Fill in member details
3. Upload CSV file

4. Review and confirm

Role Assignment

Available Roles:

- |— Instructor
 - | |— Can create courses
 - | |— Can manage students
 - | |— Can view analytics
- |— Student
 - | |— Can enroll in courses
 - | |— Can complete assessments
 - | |— Can earn certificates

Training Tracks

Training tracks group related courses into learning paths.

Creating a Track

1. Go to **Tracks** → **Create Track**
2. Enter track details:
3. Name and description
4. Target audience
5. Duration estimate
6. Add courses:
7. Select from organization courses
8. Set sequence order
9. Define prerequisites
10. Publish track

Projects & Cohorts

Projects organize training for specific groups.

Creating a Project

1. Define Project

2. Name and objectives

3. Start and end dates

4. Success criteria

5. Create Cohorts

6. Group students

7. Assign instructors

8. Set schedules

9. Assign Content

10. Select tracks or courses

11. Set deadlines

12. Configure completion requirements

Integrations

Zoom Integration

1. Go to **Settings** → **Integrations** → **Zoom**

2. Click "Connect Zoom Account"

3. Authorize access

4. Configure room defaults

Features:

- Auto-create meeting rooms
- Sync participant lists
- Record sessions

Slack Integration

1. Go to **Settings** → **Integrations** → **Slack**

2. Click "Add to Slack"
3. Select workspace
4. Choose notification channels

Notification Types:

- Course enrollments
- Completion milestones
- Deadline reminders
- Announcements

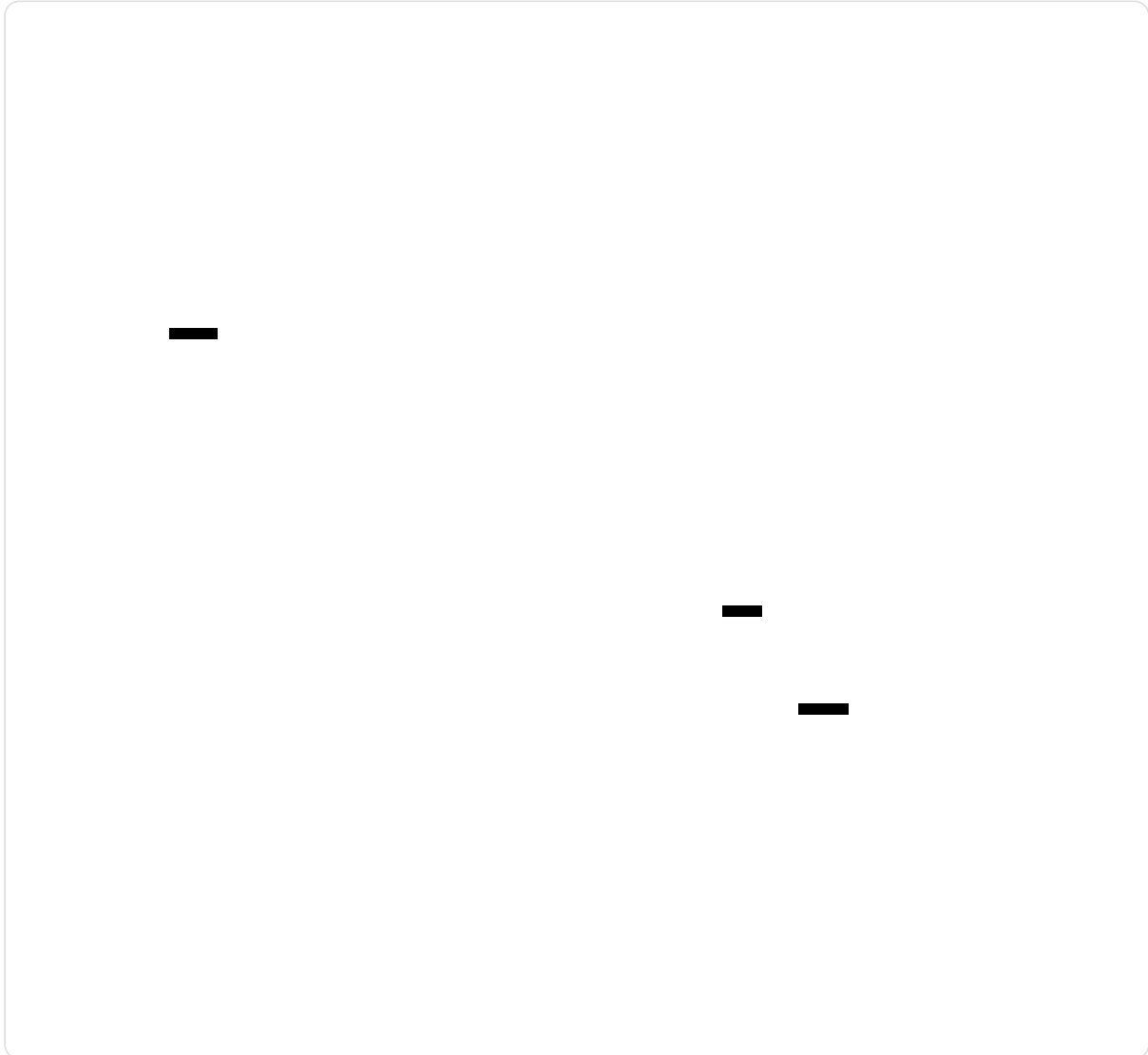
Organization Analytics

Key metrics available:

Metric	Description
Active Users	Users logged in (7/30 day)
Course Completions	Total completions
Engagement Rate	Active learning time
Compliance Rate	Mandatory training completion

Site Admin Guide

Site Admin Workflow



Platform Overview Dashboard

The site admin dashboard provides:

- **System Health:** All services status
- **Platform Metrics:** Users, orgs, courses
- **Critical Alerts:** Issues requiring attention
- **Quick Actions:** Common admin tasks

Organization Management

Creating Organizations

1. Go to **Organizations** → **Create**
2. Enter organization details
3. Set resource limits
4. Assign initial admin
5. Activate organization

Managing Organizations

Action	Description
Edit	Modify settings
Suspend	Temporarily disable
Activate	Re-enable suspended org
Delete	Permanent removal

Global User Management

User Search

Search across all organizations by:

- Username or email
- Organization
- Role
- Status

User Actions

- View profile and activity
- Reset password
- Change role

- Transfer between orgs
- Disable/enable account

System Configuration

Platform Settings

General:

- Platform name
- Contact email
- Support URL

Features:

- Registration enabled
- Guest access
- AI features
- Lab environments

Limits:

- Max organizations
- Users per org
- Storage per org
- Concurrent labs

Security Settings

- Password policies
- Session timeouts
- 2FA requirements
- API rate limits

System Monitoring

Service Status

Monitor all microservices:

- Health status
- Response times

- Error rates
- Resource usage

Log Management

Access logs for:

- Application events
- Security events
- Error tracking
- Audit trail

Demo Management

Creating Demo Data

1. Go to **Demo** → **Create Demo**
2. Select template (Small, Medium, Large)
3. Configure options
4. Generate data

Demo Reset

Reset demo environment to initial state:

1. Go to **Demo** → **Reset**
2. Confirm reset
3. Wait for completion

AI Assistant Features

Overview

The AI Assistant is available platform-wide to help with:

- **Navigation Help:** Finding features and pages
- **Learning Assistance:** Explaining concepts

- **Technical Support:** Troubleshooting issues
- **Content Creation:** Generating course materials

Accessing the AI Assistant

Click the AI bubble icon in the bottom-right corner of any page.

Conversation Features

Natural Language Queries

Ask questions naturally:

- "How do I create a new course?"
- "What are my upcoming deadlines?"
- "Explain Python list comprehensions"

Context Awareness

The assistant knows:

- Your current page
- Your role and permissions
- Your enrolled courses
- Platform features

Role-Specific Assistance

Role	AI Capabilities
Student	Study help, quiz prep, lab hints
Instructor	Content ideas, grading help, analytics
Org Admin	Setup guidance, reporting, integrations
Site Admin	System monitoring, troubleshooting

RAG-Enhanced Responses

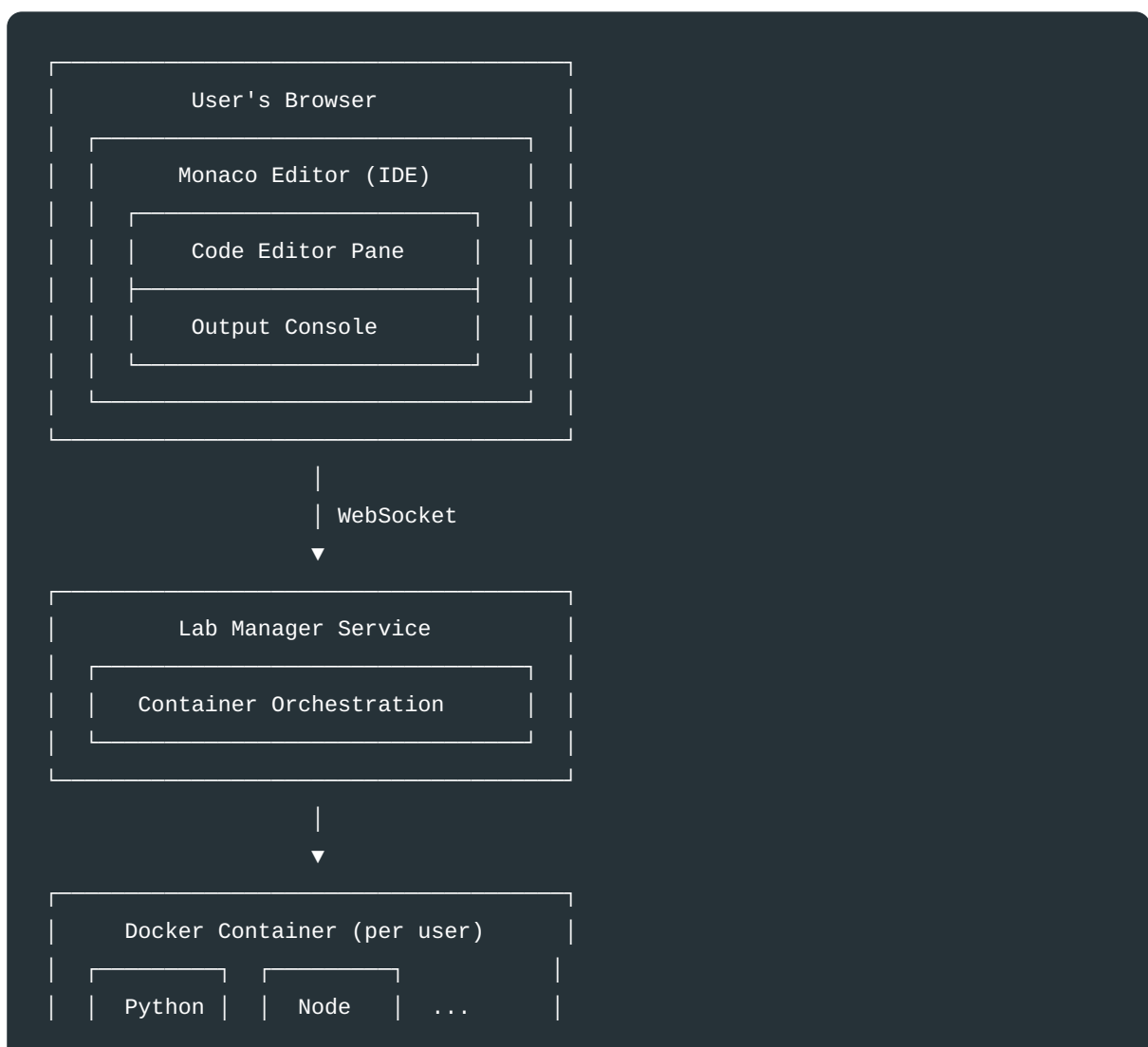
The assistant uses Retrieval-Augmented Generation (RAG) to provide:

- Course-specific answers
 - Platform documentation
 - Best practices
 - Troubleshooting guides
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Lab Environments

Architecture

Lab environments use Docker containers for isolation:





Supported Languages

Language	Version	Features
Python	3.11	Full standard library
JavaScript	Node 18	npm packages
Java	17	JDK + common libs
Go	1.21	Standard library
Rust	1.70	Cargo support

Lab Best Practices

For Students:

1. Save work frequently (auto-save enabled)
2. Test incrementally
3. Read error messages carefully
4. Use AI assistant for hints

For Instructors:

1. Provide clear instructions
2. Include starter code
3. Create comprehensive tests
4. Set appropriate time limits

Technical Reference

API Overview

The platform exposes RESTful APIs for all services:

Base URL: `https://your-domain.com/api/v1`

Authentication:

Header: `Authorization: Bearer <token>`

Common Endpoints:

GET	<code>/users/me</code>	- Current user
GET	<code>/courses</code>	- List courses
POST	<code>/courses</code>	- Create course
GET	<code>/courses/{id}</code>	- Get course
PUT	<code>/courses/{id}</code>	- Update course

WebSocket Connections

Real-time features use WebSocket:

```
// AI Assistant
ws://domain/ws/ai-assistant

// Lab Environment
ws://domain/ws/lab/{labId}
```

Database Schema

Key entities:

```
users
├─ id (UUID)
├─ username
├─ email
├─ role
└─ organization_id

courses
├─ id (UUID)
├─ title
├─ description
├─ instructor_id
└─ organization_id

enrollments
├─ user_id
└─ course_id
```

Troubleshooting

Common Issues

Login Problems

Issue	Solution
Forgot password	Use "Forgot Password" link
Account locked	Contact organization admin
SSO not working	Check with IT department

Course Access Issues

Issue	Solution
Can't see course	Verify enrollment
Content not loading	Clear browser cache
Progress not saving	Check internet connection

Lab Environment Issues

Issue	Solution
Lab won't start	Refresh page, try again
Code not running	Check syntax errors
Timeout errors	Optimize code efficiency
Lost work	Check auto-save history

Getting Help

1. **AI Assistant:** Click the AI bubble for instant help
 2. **FAQ:** Check the Help section
 3. **Support:** Contact your organization admin
 4. **Technical Support:** support@coursecreator.com
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Glossary

Term	Definition
Cohort	Group of students taking a course together
Course	Collection of learning modules
Enrollment	Student registration in a course
Lab	Interactive coding environment
Module	Section within a course
Organization	Company or institution using the platform
RAG	Retrieval-Augmented Generation (AI technique)
Track	Sequence of related courses
RBAC	Role-Based Access Control

Appendix

Keyboard Shortcuts

Shortcut	Action
Ctrl/Cmd + K	Open search
Escape	Close modal/AI Assistant
?	Show help
N	Next lesson (in course)
P	Previous lesson (in course)

Browser Compatibility

Browser	Minimum Version
Chrome	90+
Firefox	88+
Safari	14+
Edge	90+

For the latest documentation and updates, visit the platform help center or contact your administrator.

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